Event ends in 1 day 4 hours.

## Overview

In this section, you'll install and configure the **AWS Transform Custom CLI** (`atx`) on your workshop environment. The ATX CLI is the primary tool for executing .NET modernization transformations.

**Reference**
Official documentation: Getting Started with AWS Transform Custom

## Prerequisites

Before installing the ATX CLI, verify the following are available in your environment:

1. Check Node.js version (20 or later required)

```
node --version
```

Expected result

```
v20.20.2
```

2. Check Git is installed

```
git --version
```

Expected result

```
git version 2.53.0
```

3. Check .NET SDK is installed

```
dotnet --version
```

Expected result

```
8.0.126
```

4. Check Docker is installed

```
docker -v
```

Expected result

```
Docker version 29.4.0, build 9d7ad9f
```

## EXERCISE 1: INSTALL THE AWS TRANSFORM CLI

1. Run the installation script

```
curl -fsSL https://transform-cli.awsstatic.com/install.sh | bash
```

2. Verify the installation

```
atx --version
```

Expected result

```
Version: 1.7.1
```

**Troubleshooting**
If `atx` is not found, restart your terminal or run `source ~/.bashrc` to reload your PATH.

## EXERCISE 2: CONFIGURE AUTHENTICATION

AWS Transform Custom requires AWS credentials to authenticate with the service.

1. Verify your AWS credentials are configured

```
aws sts get-caller-identity
```

Expected result

```
{
    "UserId": "AROAXXXXXXXXXXXX:i-0XXXXXXXXXXXXXXXX",
    "Account": "123456789012",
    "Arn": "arn:aws:sts::123456789012:assumed-role/vscode-server-VSCodeInstanceRole-XXXXXXXXXXXX/i-0XXXXXXXXXXXXXXXX"
}
```

2. Set the AWS region (ATX Custom is available in `us-east-1` and `eu-central-1`)

```
export AWS_REGION=us-east-1
```

3. Verify the region is set

```
echo $AWS_REGION
```

Expected result

```
us-east-1
```

**IAM Permissions**
Your IAM role needs the **AWSTransformCustomFullAccess** managed policy (or at minimum **AWSTransformCustomExecuteTransformations**) to run transformations. In the workshop environment, this is already configured for you.

## EXERCISE 3: VERIFY ATX CLI IS WORKING

1. List available AWS-managed transformation definitions

```
atx custom def list
```

Expected result

```
Listing available transformation definitions...

Found 13 transformations:
- 13 AWS Managed transformations

 AWS Managed Transformations:
┌──────────────────────────────┬──────────────────────────────────────────────┬──────────┐
│ Transformation Name          │ Description                                  │ Version  │
├──────────────────────────────┼──────────────────────────────────────────────┼──────────┤
│ AWS/java-aws-sdk-v1-to-v2    │ Upgrade the AWS SDK from V1 to V2 for Java ...  │2025-10-16│
│ AWS/nodejs-aws-sdk-v2-to-v3  │ Upgrade Node.js applications from AWS SDK v2 to v3 │2025-10-16│
│ AWS/python-boto2-to-boto3    │ Migrate Python applications from boto2 to boto3 ... │2025-10-16│
│ AWS/python-version-upgrade   │ Migrate Python projects from Python 3.8/3.9 to ... │2025-10-01│
│ AWS/nodejs-version-upgrade   │ Upgrade NodeJS applications from any source ...  │2026-02-20│
│ AWS/java-version-upgrade     │ Upgrade Java applications using any build system ...│2025-10-01│
│ ...                          │                                              │          │
└──────────────────────────────┴──────────────────────────────────────────────┴──────────┘
```

2. Start an interactive ATX session to confirm connectivity

```
atx
```

Expected result



```
┌──────────────────────────────────────────────────────────────────────────────┐
│                            Region: us-east-1                                 │
└──────────────────────────────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────────────────────────────┐
│                              Trusted Tools                                   │
│                                                                              │
│ Built-in Trusted Tools                                                       │
│   Configuration: Built-in read-only tools                                    │
│   • file_read                                                                │
│   • get_transformation_from_registry                                         │
│   • list_available_transformations_from_registry                             │
│   • document_manager (add operations only)                                   │
│   • editor (view and find_line operations only)                              │
└──────────────────────────────────────────────────────────────────────────────┘

Welcome to AWS Transform. You can discover, create, and execute transformations
(AWS-managed or custom ones published to your registry). How can I help?

>
```

Type `/quit` or press `Ctrl+C` to quit.

```
/quit
```

## Key ATX CLI Commands Reference

Here's a quick reference of the commands you'll use throughout this lab:

| Command | Description |
|---|---|
| `atx` | Start an interactive conversation with the ATX agent |
| `atx --resume` | Resume the most recent conversation |
| `atx custom def list` | List available transformation definitions |
| `atx custom def exec -n <name> -p <path>` | Execute a transformation on a codebase |
| `atx custom def exec -n <name> -p <path> -x -t` | Execute non-interactively, trusting all tools |
| `atx custom def exec -n <name> -p <path> -c "<cmd>"` | Execute with a specific build command |

**Tip**
Use `atx custom def exec --help` to see all available options for running transformations.

You're all set! Let's move on to the next section where we'll prepare the .NET application for modernization.